

Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

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Abstract

Fixed task benchmarks expose important capabilities, but their rankings conflict, their accuracy scores discard most token-level evidence, and a single score cannot separate a stronger model from one trained on more of the benchmark’s subject matter: two models in our cohort sit 3.4% apart in adapted bits per byte and 2.0 points apart on HellaSwag, yet 54.6 points apart on GSM8K. Predictive cross-entropy uses every target token, yet ordinary perplexity cannot compare models with different tokenizers. Bits per byte (BPB) fixes the denominator; the open question is what to measure it on. We study that question directly, evaluating 11 base models (0.5B–35B parameters) on four fresh corpora—news, Reddit, Hacker News, and arXiv mathematics—both zero-shot and after an identical LoRA adaptation, and comparing all 24 resulting rankings against MMLU-Pro, HellaSwag, and GSM8K.

The result is not that adaptation improves the ranking. It is that adaptation *steers* it. Zero-shot news BPB already selects the GSM8K and MMLU-Pro leader outright, with zero selection regret. Adapting on general text moves the ranking toward conditional continuation—HellaSwag agreement rises from $\rho = 0.718$ to 0.982—and away from the generative tasks, where agreement falls to $\rho = 0.727$ on GSM8K and 0.764 on MMLU-Pro and the top pick now costs 2.3 and 4.2 accuracy points against the oracle. The same pattern holds on Reddit and Hacker News. Adapting instead on contamination-screened mathematical prose—not on GSM8K questions—reverses it: GSM8K agreement reaches $\rho = 0.936$, the best pairwise accuracy in the study (50 of 55 model pairs), and the top pick returns to the oracle, while HellaSwag agreement falls to 0.818. Across four adaptation corpora, the only one that beats doing nothing on GSM8K is the topically matched one.

Reported as selection accuracy over model pairs, with intervals from a cluster bootstrap over the eleven models, the selection claim holds in 23 of the 24 (corpus, tier, benchmark) cells. The contrast *between* corpora does not separate from zero at this cohort size on any single pair of them, which is why the steering claim rests on the four-corpus pattern rather than on one comparison. Because the mechanism we propose is a family-level one, we also resample by publisher rather than by model: the direction of the steering effect survives that stricter clustering—not one of 20,000 resamples reverses it—while its magnitude does not, and a leave-one-family-out jackknife shows the magnitude concentrated in a single family.

We therefore describe base-model selection as a cost–fidelity frontier rather than a single score: zero-shot BPB is free and a strong general-purpose selector, while adaptation costs compute and buys alignment with whichever capability the adaptation corpus exercises. We position the result as a second, unpreparable axis to read *alongside* a benchmark scorecard rather than as a replacement for it, and we bound its value honestly: over a cohort spanning $70\times$ in parameters model size selects as well, and the metric earns its cost only among candidates of comparable size, where size is a coin flip (0.545) and domain-matched adapted BPB reaches 0.818. We report selection regret and pairwise accuracy alongside rank correlation, and we release code, a versioned protocol, dataset fingerprints, raw harness output, and every intermediate artifact.

1 Introduction

For decades, language models were compared by predictive cross-entropy and its exponentiated form, perplexity. The metric was broad and statistically efficient—every next token contributed evidence—but token-level normalization made scores incomparable across vocabularies. Evaluation then shifted toward fixed task suites. MMLU-Pro, HellaSwag, and GSM8K are useful precisely because they isolate different capabilities [24, 25, 26]; the same specialization also makes their rankings disagree. Once benchmark questions become development targets or enter pre-training corpora, a high score may additionally reflect task preparation or overlap rather than transferable base-model quality [15, 20].

The difficulty is concrete rather than rhetorical, and it is visible inside this cohort. Llama-3.2-1B and Qwen-2.5-1.5B sit at adjacent positions in our adapted news BPB ranking, 3.3% apart, and 2.0 points apart on HellaSwag. On GSM8K they are 54.6 points apart. Qwen-3.5-4B and Llama-3.2-1B are 2.9% apart on the same ranking and 73.9 points apart on GSM8K. Two models can therefore model fresh English almost identically well and differ nine-fold on a benchmark, and no published score distinguishes “this model is stronger” from “this model was trained on more of this”. Both are legitimate design choices; the point is that the score cannot tell a reader which one it is reporting, and the pre-training mixtures that would settle it are not disclosed. A metric computed on text the evaluator owns, dated after every candidate shipped, cannot have been prepared for—which is a different guarantee from being a better measure of quality, and we are careful throughout to claim only the former.

Recent evidence makes the return to loss timely. Cross-entropy can have higher signal and lower noise than accuracy [18], and train-before-test evaluations show that light, task-specific preparation substantially harmonizes benchmark rankings and restores their connection to perplexity [20]. The unresolved practical question is how to make that connection useful when choosing among off-the-shelf base models with different tokenizers for a new domain.

We answer with a controlled *adapt-then-measure* protocol on evaluator-owned text. Bits per byte (BPB) places every model on one physical denominator. A recent target-domain snapshot limits staleness; internal deduplication and model-based risk signals trigger conservative quarantine before training or scoring; and a fixed LoRA configuration measures candidates under the same data and update budget. The endpoint, adapted BPB, answers “which base model best models this domain after equal preparation?” The reduction from zero-shot to adapted BPB answers a different question: “how much calibration did this corpus require?” Neither number is presented as a substitute for a task benchmark when the deployment task is already known.

The finding that organizes this paper is what happens when both endpoints are measured against external benchmarks instead of only the adapted one. Adaptation does not move the ranking uniformly closer to any external notion of model quality; it rotates the ranking toward whatever the adaptation corpus exercises. That makes the choice of corpus the substantive decision in the protocol, and it makes the zero-shot ranking—which is free—a serious baseline rather than an intermediate quantity. We consequently report both tiers everywhere, and we judge each of them by what it costs a practitioner who acts on it.

Our contributions are: (1) a tokenizer-agnostic, fresh-data protocol for comparing adaptable base models, with a versioned estimand and auditable artifacts; (2) the complete $4 \times 2 \times 3$ alignment matrix—four adaptation corpora, zero-shot and adapted, against three benchmarks—which makes the zero-shot baseline and the non-matched-corpus control explicit rather than assumed; (3) selection regret and pairwise accuracy as reporting statistics that a rank correlation over 11 models cannot support

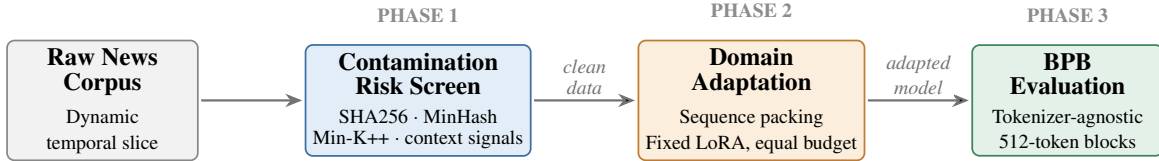


Figure 1: The standardized adapt-then-measure pipeline. Deduplication and contamination-risk screening (Phase 1) precede controlled adaptation (Phase 2) and marker-masked BPB evaluation (Phase 3).

on its own; (4) evidence that the adaptation corpus steers which capability the ranking tracks, demonstrated by a math-domain intervention that is checked against three general-text controls rather than one; (5) token- and context-level diagnostics of what the adapters learn, including the finding that a fixed update budget measures calibration *speed* rather than attainable fit; and (6) a public implementation with benchmark manifests, dataset fingerprints, raw harness output, and a validated leaderboard. Code and artifacts are available at <https://github.com/DavidAfonsoValente/entropy-bench>.

2 Related Work

From perplexity to reliable loss evaluation. Language modeling was originally grounded in byte- and character-level entropy [1], while token-level perplexity depends on each model’s vocabulary and segmentation [2, 4]. Normalizing log-likelihood by UTF-8 bytes restores a common unit: The Pile used BPB for cross-tokenizer comparison [3], and Paloma standardized broad, domain-resolved measurement of language-model fit [5]. Heineman et al. evaluate 465 models over 30 benchmarks and show that loss-based metrics can offer better signal, noise, and scaling-law reliability than accuracy [18]. Our contribution is therefore not BPB itself; it is the operational protocol around the metric for comparing heterogeneous, off-the-shelf base models on new target data.

Why task rankings disagree. MMLU-Pro, HellaSwag, and GSM8K measure knowledge, conditional continuation, and mathematical problem solving under distinct elicitation and scoring rules [24, 25, 26]. Such scores can be indispensable for a known deployment task, but accuracy is discrete and benchmark-specific. Observational scaling work finds that a low-dimensional capability structure explains much public-model performance [21], while controlled model families exhibit a smooth relationship between validation loss and average downstream error [22]. We test that relationship across families and report both alignment and systematic residuals rather than collapsing the tasks into an uninterpreted macro-average.

Which corpus the loss is measured on. That the agreement between predictive loss and benchmark performance depends on *where* the loss is measured is not new. Thrush et al. show that per-domain perplexity correlates with benchmark performance to systematically different degrees across web domains, and turn that dependence into a pretraining-data selection rule [19]. Their estimand is the complement of ours: they hold the benchmark suite fixed and choose data to *train* one strong model, using a pool of existing models to estimate which domains carry signal. We hold the candidate pool fixed, adapt every member on the same corpus, and ask how the resulting *selection ranking* moves against several benchmarks at once. The domain-dependence they document is the premise our steering result builds on; what we add is that the dependence can be actuated—the corpus is a control input for which capability the ranking tracks, with the matched corpus improving one benchmark’s

agreement while degrading another’s, and three general-text controls failing to do either.

Equal preparation. Dynamic evaluation updates a language model on recent test history to improve predictive likelihood [27, 28]; domain-adaptive pre-training improves fit to a target distribution [29]. Closest to our model-selection motivation, Zhang et al.’s *train-before-test* gives 61 models identical supervised, benchmark-specific preparation across 24 tasks: average cross-benchmark Kendall agreement rises from 0.52 to 0.76, and pre-adaptation BPB predicts post-adaptation task potential for base models [20]. Their result motivates our design but the estimand differs. They rank post-fine-tuning task accuracy on public labeled benchmarks; we rank held-out language-model fit after unsupervised adaptation to a fresh target-domain corpus. We also standardize the byte denominator, screen contamination risk, and test rank transfer across raw-text domains.

Contamination controls. Models can memorize duplicated pre-training text [8, 9]; deduplication and contamination analysis are therefore necessary [10, 15]. Reference-free membership signals such as Min-K% and Min-K++ are useful screens [11, 12], but no black-box score proves that a sample occurred in a proprietary training set. We consequently use soft signals only for conservative quarantine and report them as risk indicators, not verified leaks.

3 The Evaluation Pipeline

The pipeline (Figure 1) is chronological: corpus construction and screening are completed before any sample can influence adaptation or scoring.

3.1 Phase 1: Deduplication and Contamination-Risk Screening

We first enforce exact split disjointness and remove exact and MinHash/LSH near-duplicates. For models whose pre-training corpora are unavailable, we then apply three reference-free screens:

- **Min-K++:** average the normalized likelihood of the lowest-probability $K = 20\%$ target tokens, using the model’s probability-weighted vocabulary mean and variance at each position; unusually high within-split outliers become candidates.
- **Context sensitivity:** compare target NLL with and without randomly sampled same-corpus prefix documents; unusually large confidence gains add a soft risk signal.
- **Perturbation preference:** for already flagged candidates, compare the original sequence with two word-swap/placeholder perturbations. Preference for the original triggers conservative quarantine, but is not treated as proof of memorization because fluency alone can produce the same outcome.

The screen runs per model and the union of candidate indices is quarantined globally. Thus every model receives the same retained corpus. This policy reduces overlap risk without making an unverifiable claim about membership in proprietary pre-training data.

This screen is applied to the news corpus. The three later corpora are protected by a stronger and simpler property instead: every document in them was published after every model in the cohort was released, so no detector is needed to establish that they were not trained on (Table 1). We regard freshness, not the detector stack, as the load-bearing contamination defence in this work, and we recommend it over a screen wherever a target domain permits a recent snapshot.

3.2 Phase 2: Controlled Domain Adaptation

The retained corpus is packed into contiguous $L=512$ -token blocks along document boundaries. Every headline model–corpus cell uses LoRA rank 16, $\alpha = 32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, all-linear targets, and 250 update steps. This fixes the data and update budget, although larger models still require more FLOPs and have more trainable LoRA parameters.

Why a fixed configuration rather than a per-model search. An earlier version of this experiment selected each model’s adapter by a multi-fidelity Optuna sweep. We abandoned that design because its search budget was confounded with model size: within a fixed wall-clock allocation the two largest models completed four trials each, while the 1B models completed 17 and 18 (results/legacy_sweep/*/trials.csv). A search that is systematically deeper for small models hands them a better-tuned adapter and biases adapted BPB by parameter count, by an amount that cannot be estimated after the fact. One shared configuration removes that confound outright; it is tuned to no model’s advantage, and it is the reason the leaderboard is comparable at all. The cost is that a fixed configuration is not equally well suited to every architecture, which Section 6 states as a limitation and Section 6.1 proposes to resolve.

Separately, the mechanistic diagnostics in Section 5.2 use three adapters selected by that exploratory sweep; those runs explain the source of adaptation gains but do not determine the public ranking.

3.3 Phase 3: Tokenizer-Agnostic BPB Evaluation

Cross-entropy loss ($\mathcal{L}_{\text{nats}}$) aggregated over the held-out corpus is unified onto byte-space using the UTF-8 byte-length ($|B|$) of the source text:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln(2)} \times \frac{|Tokens|}{|B|} \quad (1)$$

Zero-shot and adapted BPB are both measured on the held-out test split. Loss is accumulated over scored tokens and divided by the exact UTF-8 byte count represented by those targets. The packed test stream uses independent, non-overlapping 512-token blocks, so context resets at every cut. Injected document-start markers remain available as context but are always masked as prediction targets in the headline protocol. This point matters: an older exploratory run scored those markers; all core tables, correlations, the manifest, and the leaderboard use the corrected marker-masked rerun.

4 Experimental Setup

Models. We evaluate 11 base checkpoints spanning dense Transformers, one sparse MoE, and one Liquid architecture (0.5B–35B parameters); the cohort is listed in Tables 2 and 5.

Data. The primary snapshot is Google News dated 8 June 2026 (119,054 articles, ~ 361 MB of text). Reddit and Hacker News provide the transfer check in Table 1. Every corpus uses an 80/10/10 train/validation/test split (seed 42) with zero ID and normalized-text intersection across splits.

Static benchmarks. For Section 5.4, every base model is scored with *lm-evaluation-harness* 0.4.12 and vLLM 0.25.1 (bf16, greedy generation, tokenizer-native BOS). MMLU-Pro uses 5-shot chain-of-thought and exact match on one stratified 1,000-question subset (seed 42); HellaSwag uses 10-shot length-normalized multiple-choice accuracy on all 10,042 validation examples; GSM8K uses 5-shot chain-of-thought and flexible numeric extraction on all 1,319 test examples. These are directly observed scores under one harness, not vendor-reported model-card values: the aggregate `results_*.json` and per-example `samples_*.jsonl` for all 11 models on all three benchmarks are published under `results/gsm8k/`, `results/hellaswag/`, and `results/mmlu_pro_1k/`, so every number in Table 5 recomputes directly from the raw harness output (`tools/verify_paper_numbers.py`).

Descriptive uncertainty and ranks. We orient every rank so 1 is best, compute Pearson correlation on score values, Spearman correlation on ranks, and Kendall agreement on pairwise order. Wilson 95% intervals show test-set sampling uncertainty for accuracy. They do not include run-to-run, prompt, subset-selection, or model-family uncertainty; the cohort is purposive, so all correlations are descriptive rather than population estimates.

Hardware. Adaptation ran under Slurm on the Leonardo (CINECA) GPU partition, 3 nodes $\times$ 4 NVIDIA A100 64 GB; the static-benchmark suite ran separately on A100 GPUs.

5 Results and Discussion

5.1 Core Result: Adapted Rankings Transfer Across Domains

We first test the claim needed for practical model selection: after the same light adaptation, does the ordering identify broadly strong base models rather than one news-specific winner? We prepare three corpora with the same grouping and cleaning recipe—exact deduplication, Gopher quality heuristics, and MinHash/LSH near-duplicate filtering at Jaccard 0.60—and run all 11 models on every corpus. Table 1 also records how each corpus is protected against contamination, which is not uniform across the four and should not be read from Figure 1 as if it were.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K++ / context / perturbation union
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv math. *, 9 Jun–20 Aug 2026 [‡]	3,207	3,207	—	15 MB	Post-release scrape + GSM8K shingle screen

Table 1: Corpora. All four receive identical exact-deduplication, Gopher quality filtering, and MinHash/LSH near-duplicate removal at Jaccard 0.60. They differ in *contamination* control, and deliberately so: the reference-free screen of Section 3 was run on the news corpus, whose 8 June 2026 snapshot predates part of the cohort’s release window, while the other three were scraped after every model in the cohort shipped and are protected by construction rather than by a detector. Reddit and Hacker News comments are concatenated into document-length groups. [†]The Reddit source contains four available days in the stated range. [‡]Used only for the intervention in Section 5.5, not for the 33-cell transfer experiment.

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LiquidAI-LFM2.5	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 2: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

Adaptation changes the choice. Zero-shot and adapted rankings are only moderately aligned: Spearman $\rho = 0.655$ on news, 0.609 on Reddit, and 0.682 on Hacker News, with maximum movements of seven, seven, and five places. Gemma-4-31B moves from fourth, fifth, and sixth zero-shot to first adapted; Gemma-4-12B moves from ninth in every zero-shot ranking to second, second, and fourth. Adapted BPB is therefore not a rescaling of ordinary held-out loss; it estimates fit after equal target-domain preparation.

The adapted order transfers. News and Reddit agree on nine of eleven exact ranks (Spearman $\rho = 0.991$; maximum displacement one). News versus Hacker News yields $\rho = 0.955$ and Reddit versus Hacker News $\rho = 0.964$, with maximum displacement two. Gemma-4-31B ranks first on all three corpora. Figure 2 shows that the result is not only ordinal: Pearson score correlations are 0.992, 0.977, and 0.990.

Rank changes are near-tie changes. The three-model mid-size tier spans 0.0259 BPB on news but only 0.0089 on Hacker News; the two-model small tier spans 0.0258 and 0.0088. On Hacker News, Ministral-3-14B (0.835), Qwen-3.5-35B-MoE (0.836), and Gemma-4-12B (0.844) are separated by 0.009 BPB. A target-domain run should resolve choices inside such a tier; the cross-domain result supports the broader tier ordering, not an immutable total order.

Difficulty and order are distinct. Every model has its lowest BPB on news and highest on Reddit. Gemma-4-31B moves from 0.624 on news to 0.913 on Reddit, a 46% increase, yet remains first. Absolute BPB describes domain difficulty and attainable fit; relative rank describes which candidate is strongest under the shared adaptation protocol.

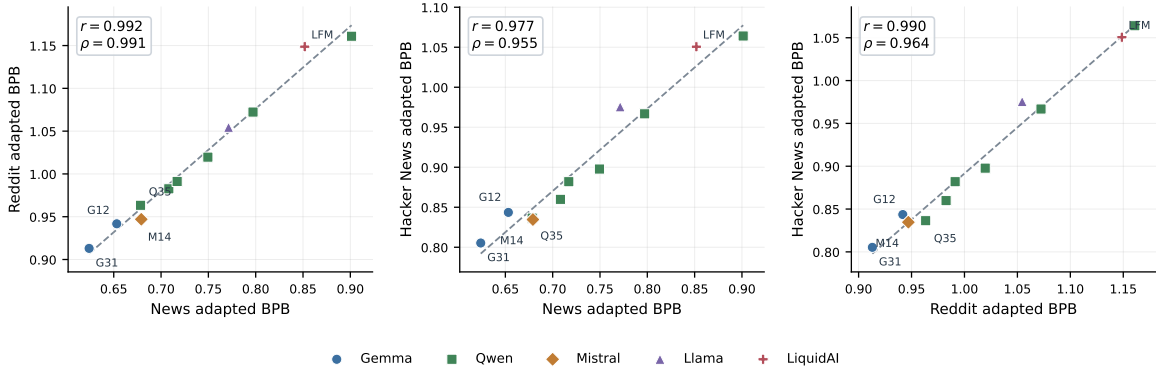


Figure 2: Pairwise adapted-BPB agreement across domains. Absolute difficulty changes, but points remain close to a monotone shared ordering. G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

5.2 Mechanistic Diagnostics: What Adaptation Changes

To interpret the reduction, we score held-out blocks under each base model and an exploratory hyperparameter-selected LoRA adapter, then decompose the per-token log-probability gain. These three models span the observed range. This diagnostic run predates the marker-masking correction and is reported only to localize gain; it does not supply any headline BPB or leaderboard rank.

Model	Loss (nats/token)			Share of the gain		
	base	adapted	Red.	marker	next 16	remainder
Gemma-4-12B	2.838	2.015	29.0%	3.6%	5.0%	91.3%
LiquidAI-LFM2.5	4.298	2.728	36.5%	1.1%	1.9%	97.0%
Qwen-2.5-1.5B	2.737	2.647	3.3%	0.0%	0.0%	100%

Table 3: Exploratory token-level decomposition over 4,000 test blocks. *marker* is an injected document-start token, *next 16* is the post-boundary window, and *remainder* is every other position. Headline evaluations mask the marker target.

Surface-form calibration. Gemma-4-12B’s largest movers are `<bos>`, punctuation, paragraph breaks, and common function words; LFM-2.5 shows the same pattern. Across the top 200 gaining tokens, formatting, punctuation, and function words account for 83% of the gain for both models, while content words account for 5.2% for Gemma and 14.0% for LFM. Qwen-2.5-1.5B begins closer to the corpus distribution and allocates more of its smaller gain to dates and figures. For Gemma, the marker and its next 16 tokens contribute $3.6\% + 5.0\% = 8.6\%$ of this exploratory gain. These observations explain the three selected adapters; they are not assumed to identify a universal mechanism for all 11 models.

Context reveals steady-state quality. Figure 3 resolves loss and gain by position in the 512-token block and normalizes every bucket to UTF-8 bytes. Over the first eight positions, Gemma-4-12B averages 2.566 bits/byte and Qwen-2.5-1.5B averages 1.490; by positions 256–510, Gemma reaches 0.717 and Qwen 0.773. The crossing therefore remains after byte normalization and is not explained by vocabulary size alone. Once context is available, the 12B model shows its stronger steady-state language modelling. The opening 32 positions account for 23.0% of Gemma’s total gain and 12.9% of Qwen’s, making context recovery a clear component of the calibration signal.

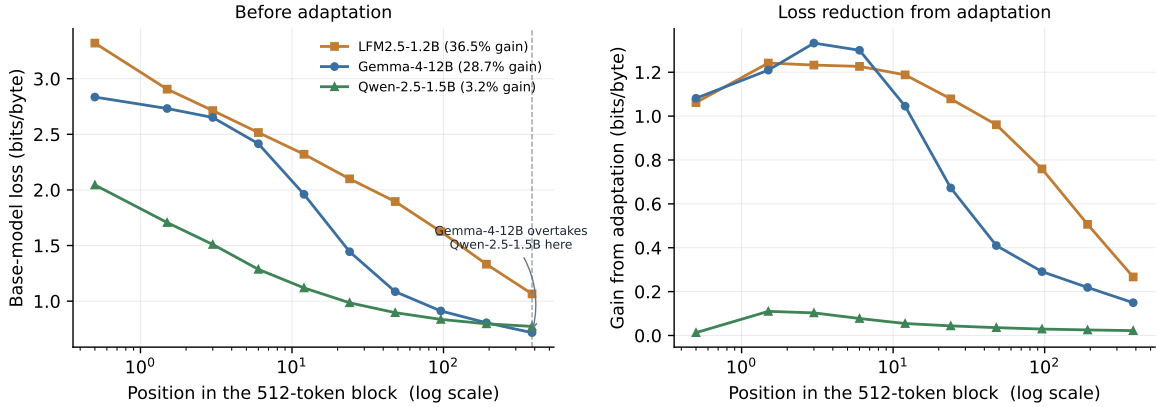


Figure 3: Exploratory base loss and adaptation gain by position in a packed 512-token block, in bits per UTF-8 byte. Positive gain is the loss reduction. The late-context crossing between Gemma-4-12B and Qwen-2.5-1.5B persists after byte normalization.

The 512-token adaptation transfers to longer inference contexts. We select 64 fixed held-out target windows and vary only the preceding context from 16 to 2,048 tokens, scoring the same target tokens at every length with exact byte normalization. Figure 4 shows that Gemma’s base BPB continues to recover beyond the training window, from 0.6465 at 512 tokens to 0.6054 at 2,048. Its adapted BPB is effectively unchanged ($0.5473 \rightarrow 0.5474$), so the adapter trained on 512-token blocks retains a 9.6% advantage at four times that context. Qwen’s smaller adaptation advantage also remains positive at 2,048 tokens ($0.7400 \rightarrow 0.7271$, a 1.7% reduction). Thus the base model continues to benefit from longer context, especially for Gemma, while the learned adaptation remains stable outside its training length.

Interpretation. The adapted BPB level remains the selection metric. The zero-shot→adapted reduction is a calibration diagnostic; it must not be ranked as if a larger change implied a better endpoint.

5.3 Contamination-Risk Screen: What Was Removed

Table 4 reports the soft screening funnel on 1,000 sampled test sequences per model. It is a removal audit, not an estimate of each model’s memorization rate.

The near-100% preference rates are expected because the distractors contain a word swap and a placeholder; a model can prefer the fluent original without having memorized it. We therefore make no scale–memorization claim from this table. The operational guarantee is narrower and auditable: all models use the same corpus after the union of flagged indices is removed.

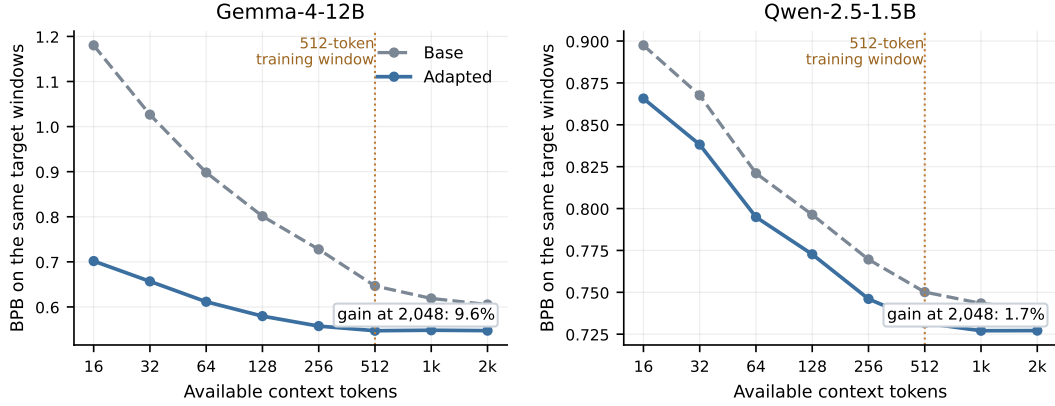


Figure 4: Fixed-target context-length diagnostic. Each point scores the same 64 held-out target windows; only preceding context changes. Both exploratory 512-trained adapters retain a positive gain through 2,048 inference tokens.

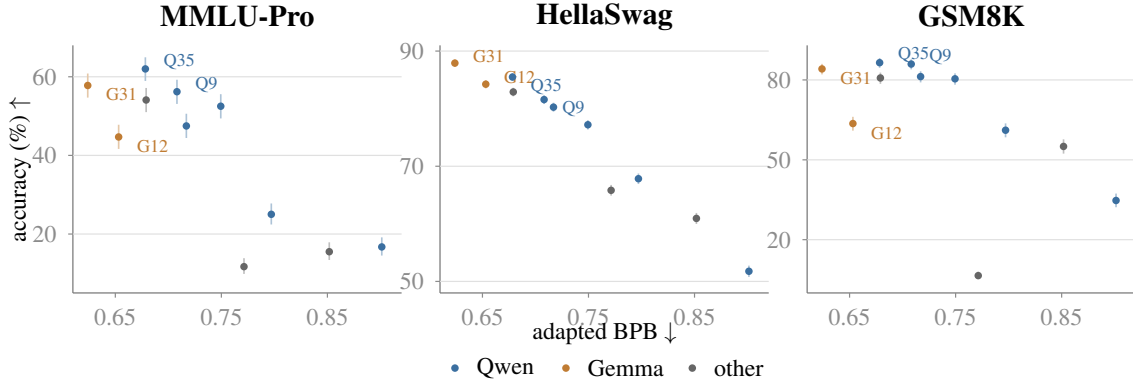


Figure 5: Benchmark accuracy versus corrected adapted-news BPB. Vertical bars are Wilson 95% intervals for finite test-set sampling only. Labels mark the four models central to the rank differences (G31/G12: Gemma; Q35/Q9: Qwen).

5.4 Correlation with Static Benchmarks

We now compare the corrected adapted-news BPB with all three benchmark accuracies. Table 5 contains the complete 11-model intersection. Parentheses give within-column rank, always oriented so 1 is best.

Adaptation does not improve the ranking; it rotates it. Table 7 contains the result that organizes the rest of this paper, and it is only visible once the zero-shot tier is reported alongside the adapted one. On news, adaptation raises HellaSwag agreement from $\rho = 0.718$ to 0.982 and simultaneously *lowers* GSM8K agreement from 0.836 to 0.727 and MMLU-Pro from 0.855 to 0.764. The same three-way movement appears on Reddit (0.655 $\rightarrow$ 0.964, 0.791 $\rightarrow$ 0.691, 0.791 $\rightarrow$ 0.736) and on Hacker News (0.627 $\rightarrow$ 0.973, 0.827 $\rightarrow$ 0.791, 0.818 $\rightarrow$ 0.845). Three independent corpora, one direction.

Table 8 states the same fact as a decision. Zero-shot news BPB selects Qwen-3.5-35B-MoE,

Model	Min-K++ flags	Context flags	Candidates	Original preferred	Quarantined
Gemma-4-31B	12	6	20	100%	20
Gemma-4-12B	11	6	11	100%	11
Ministral-3-14B	12	8	13	100%	13
Qwen-3.5-35B-MoE	6	8	11	100%	11
Qwen-3.5-9B	12	6	17	100%	17
Qwen-2.5-7B	10	6	14	100%	14
Qwen-3.5-4B	9	6	12	100%	12
Llama-3.2-1B	17	0	28	100%	28
Qwen-2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-2.5-0.5B	8	2	13	100%	13

Table 4: Conservative screening funnel on the *test split* ($N = 1000$ sampled sequences per model; all splits are screened). “Original preferred” is preference over two synthetic perturbations, not task accuracy or proof of training membership. The global union removed 80 unique candidate sequences across splits and models.

which is the GSM8K and MMLU-Pro leader—zero regret on both—at the cost of 2.4 points against Gemma-4-31B on HellaSwag. After adaptation the top pick becomes Gemma-4-31B: zero regret on HellaSwag, 2.3 and 4.2 points of regret on GSM8K and MMLU-Pro. Adaptation did not find a better model. It changed which question the ranking answers.

HellaSwag and BPB measure the same local object. The direction of that rotation is not mysterious. HellaSwag selects a continuation by length-normalized conditional log-likelihood, whereas BPB aggregates conditional log-likelihood over every target token; further language-model training on in-domain text should, and does, pull the two into near-identical order ($\rho = 0.982$: seven models keep their exact rank and none moves more than one place; Gemma-4-31B leads both at adapted BPB 0.624 and HellaSwag 87.9%). This is convergent validity between two measurements of the same local object, and it is the least surprising cell in Table 7 rather than the protocol’s strongest evidence. The generative benchmarks, which require producing and extracting an answer, move the other way.

Knowledge and elicitation. MMLU-Pro agreement remains substantial but weaker ($\rho = 0.764$). The Qwen-3.5-35B-MoE leads at 62.0% and moves from BPB rank 3 to MMLU rank 1; Qwen-3.5-9B moves from 5 to 3. Gemma-4-12B moves from BPB rank 2 to MMLU rank 7. Unlike HellaSwag’s likelihood comparison, our MMLU-Pro protocol generates 5-shot chains of thought and requires an extractable exact answer. The residual therefore mixes factual coverage, reasoning, few-shot pattern continuation, and answer formatting.

GSM8K exposes a family-specific residual. No model keeps the same rank. Every Qwen model is unchanged or improves relative to adapted news BPB; both Gemma models decline, with Gemma-4-12B moving from rank 2 to 7. The contrast is large even among small models: Qwen-2.5-1.5B reaches 61.1% GSM8K at BPB rank 9, while Llama-3.2-1B reaches 6.5% at BPB rank 8. This supports a narrow interpretation: arithmetic data, chain-of-thought preparation, and generation-format compatibility vary systematically by model family. Architecture, data mixture, token budget, and

Model	Adapted BPB ↓	MMLU-Pro ↑	HellaSwag ↑	GSM8K ↑
Gemma-4-31B	0.624 (1)	57.8 (2)	87.9 (1)	84.2 (3)
Gemma-4-12B	0.653 (2)	44.7 (7)	84.2 (3)	63.6 (7)
Qwen-3.5-35B-MoE	0.678 (3)	62.0 (1)	85.5 (2)	86.5 (1)
Ministral-3-14B	0.679 (4)	54.1 (4)	82.9 (4)	80.7 (5)
Qwen-3.5-9B	0.708 (5)	56.2 (3)	81.6 (5)	86.0 (2)
Qwen-2.5-7B	0.717 (6)	47.5 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.5 (5)	77.2 (7)	80.4 (6)
Llama-3.2-1B	0.771 (8)	11.7 (11)	65.8 (9)	6.5 (11)
Qwen-2.5-1.5B	0.797 (9)	25.0 (8)	67.8 (8)	61.1 (8)
LiquidAI-LFM2.5	0.852 (10)	15.5 (10)	60.9 (10)	55.0 (9)
Qwen-2.5-0.5B	0.901 (11)	16.7 (9)	51.8 (11)	34.7 (10)

Table 5: Directly observed benchmark accuracy in percent, with rank in parentheses. Bold marks the best point estimate in each column; it does not assert a statistically resolved lead. Adapted BPB comes from the marker-masked fixed-protocol news run.

	Pearson r	Rank ρ	Kendall τ	Exact ranks	Mean shift	Max shift
MMLU-Pro	−0.826	0.764	0.564	3/11	1.64	5
HellaSwag	−0.975	0.982	0.927	7/11	0.36	1
GSM8K	−0.614	0.727	0.564	0/11	2.00	5

Table 6: Alignment with adapted news BPB. Pearson correlates BPB with accuracy, hence negative is agreement. Spearman and Kendall compare best-to-worst order and are oriented positive. “Shift” is absolute rank displacement. Table 7 places these three rows in the context of every other corpus and both cost tiers.

release generation are confounded here, so this correlational comparison alone cannot rank those causes—which is what Section 5.5 probes with an intervention on the adaptation corpus.

Two cautions belong with this residual. First, it is a residual against the *adapted* ranking specifically: zero-shot news BPB places Qwen-3.5-35B-MoE, Ministral-3-14B, and Qwen-3.5-9B in its top three and orders 46 of 55 pairs correctly against GSM8K, so the family-specific disagreement is substantially something adaptation introduces rather than something it reveals. Second, the family claim rests on two Gemma models, which is an observation about this cohort and not yet a demonstrated family effect.

The residual is a demonstration, not an estimate. It is tempting to promote that observation into a statistic: correlate [BPB gap] against [benchmark gap] over all 55 pairs and report that BPB gaps predict HellaSwag gaps but not GSM8K gaps. The full-cohort coefficients invite it ($r = +0.909$ for HellaSwag against $r = +0.266$ for GSM8K). We do not report that contrast, because it does not survive deletion: dropping Llama-3.2-1B alone moves the GSM8K coefficient to $+0.667$ and dropping Qwen-2.5-0.5B moves it to $+0.118$, a swing larger than the coefficient’s own distance from zero, while HellaSwag stays within $[0.859, 0.940]$ under every single-model deletion. Each model sits

Corpus	Tier	GSM8K	MMLU-Pro	HellaSwag	GSM8K pairs
News	zero-shot	0.836	0.855	0.718	46/55
	adapted	0.727	0.764	0.982	43/55
Reddit	zero-shot	0.791	0.791	0.655	44/55
	adapted	0.691	0.736	0.964	42/55
Hacker News	zero-shot	0.827	0.818	0.627	45/55
	adapted	0.791	0.845	0.973	45/55
Math (arXiv)	zero-shot	0.782	0.755	0.482	44/55
	adapted	0.936	0.936	0.818	50/55

Table 7: Rank agreement (Spearman ρ) between every BPB variant and every benchmark. Four adaptation corpora $\times$ two cost tiers $\times$ three benchmarks; the two rows the earlier literature and our own previous draft reported are *News adapted* and *Math adapted*. “GSM8K pairs” counts model pairs ordered correctly out of 55. Generated by `tools/analyze_alignment_matrix.py` into `results/alignment_matrix.json`.

in ten of the fifty-five pairs, so one unusual model moves the fit a long way; this is the same leverage failure recorded for the BPB-movement correlation in Section 6.2, and it is reported here rather than avoided.

What survives is weaker in form and sufficient for the argument. Among the ten pairs within 5% of each other on adapted news BPB, the six cross-family pairs have a median GSM8K gap of 20.0 points and the four same-family pairs a median of 2.8 points: when the metric says two models from the same publisher are tied, the benchmark agrees, and when it says that about two models from different publishers, it may not. Those are descriptive counts over single-digit cells. The named pairs quoted in Section 1 are existence proofs about specific released models, so the leverage argument does not apply to them, and they are all the motivation requires. All of it regenerates via `tools/analyze_selection_divergence.py`.

Two benchmarks, not three. MMLU-Pro and GSM8K are not independent evidence in this cohort: they agree on 51 of 55 model pairs (Kendall $\tau = 0.855$). Wherever both move together in Tables 7 and 8—as they do in every row—that is one finding reported twice, not two corroborating findings. The study therefore resolves two capability axes, a generative one shared by MMLU-Pro and GSM8K and a likelihood one measured by HellaSwag.

Point-estimate leaders are not equally resolved. HellaSwag’s leader exceeds the runner-up by 2.41 percentage points (independent-binomial $z = 5.03$), and their Wilson intervals do not overlap. MMLU-Pro’s 4.2-point top gap has $z = 1.92$ and overlapping intervals; GSM8K’s top two Qwen models differ by only 0.53 points ($z = 0.40$). Boldface therefore identifies the observed maximum, not a universal or necessarily significant winner. The repo publishes the numerators, denominators, intervals, ranks, and derivation script in `results/static_benchmark_analysis.json`, and the raw harness output each of those numbers is computed from in `results/gsm8k/`, `results/hellaswag/`, and `results/mmlu_pro_1k/`.

BPB variant	Top pick	Selection regret (accuracy points)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	2.3	4.2	0.0
Reddit zero-shot	Ministral-3-14B	5.8	7.9	5.0
Reddit adapted	Gemma-4-31B	2.3	4.2	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	2.3	4.2	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 8: What it costs to act on each metric. Regret is the accuracy of the benchmark’s own best model minus the accuracy of the model this BPB variant ranks first. The oracle is Qwen-3.5-35B-MoE for GSM8K and MMLU-Pro and Gemma-4-31B for HellaSwag, so a regret of 0.0 means the metric selected the benchmark’s leader outright.

5.5 Math-Domain Adaptation Narrows the GSM8K Residual

Section 5.4 left the GSM8K residual unresolved: news-adapted BPB agrees with GSM8K rank only moderately ($\rho = 0.727$, 0 of 11 exact ranks), and we attributed this to a confound of arithmetic data, chain-of-thought preparation, and answer-format compatibility that varies by model family. That framing predicts the residual should *not* respond to more adaptation unless the adaptation corpus is itself math-domain text. We test this directly, outside the three-domain, 33-cell design of Section 5.1: we repeat the adapt-then-measure protocol, with the same fixed LoRA configuration as the other three domains, on a corpus of independent math-domain prose rather than GSM8K questions. The corpus is 3,207 arXiv math.* titles and abstracts submitted 9 June–20 August 2026—strictly after every model’s release, the same freshness argument used for Reddit and Hacker News—built with `tools/prepare_math_corpus.py` and screened against the GSM8K test split by 10-word-shingle containment before any model saw it (0 hits), using the same forbidden-corpus overlap check as the paper’s contamination audit.

Adaptation corpus	GSM8K ρ	Pairs correct	$\Delta\rho$ vs. no adaptation
<i>None</i> (zero-shot news)	0.836	46/55	—
News	0.727	43/55	−0.109
Reddit	0.691	42/55	−0.145
Hacker News	0.791	45/55	−0.045
Math (arXiv)	0.936	50/55	+0.100

Table 9: The control that makes the math result interpretable: GSM8K rank agreement under each of the four adaptation corpora, all against the same do-nothing baseline. Three general-text corpora leave GSM8K agreement at or below the zero-shot level; only the topically matched corpus exceeds it. Derivation in `results/alignment_matrix.json` (`tools/analyze_alignment_matrix.py`).

Judge it against doing nothing, not against news. It is tempting to report this result as $\rho = 0.727 \rightarrow 0.936$, and an earlier draft of this paper did. That contrast overstates the effect, because

0.727 is not a neutral starting point—it is *below* the 0.836 that zero-shot news BPB achieves with no adaptation at all. Roughly half of the apparent gain is news-adaptation damage being undone rather than domain matching creating signal. Measured against the do-nothing baseline the improvement is +0.100, and against every general-text corpus we tried it is between +0.145 and +0.245.

A reader may reasonably ask why the do-nothing baseline for a *math* corpus is zero-shot *news* BPB. It is because that is the harder test: zero-shot news BPB reaches 0.836 on GSM8K while zero-shot math BPB reaches only 0.800, so scoring the intervention against the news baseline subtracts more from it, not less. Using the corpus-matched zero-shot number would have reported +0.136 rather than +0.100. We report the more conservative of the two.

The residual narrows, and HellaSwag pays for it. What survives that correction is still the strongest selection result in the study. Math-adapted BPB reaches $\rho = 0.936$ on GSM8K with 6 of 11 exact ranks and 50 of 55 model pairs ordered correctly—the best pairwise accuracy of any variant in Table 7—and it is the only adapted variant whose top pick is the GSM8K oracle (Table 8). MMLU-Pro moves with it (0.764 $\rightarrow$ 0.936), which given the $\tau = 0.855$ redundancy between the two generative benchmarks is corroboration of the same axis rather than a second result. HellaSwag moves the other way (0.982 $\rightarrow$ 0.818), and we report it with equal prominence: this is not a free improvement but a trade of general conditional-continuation calibration for domain-specific calibration, the textbook cost of narrowing what the adaptation data covers.

The reordering is mechanistically legible. Under news-adapted BPB, Gemma-4-31B and Gemma-4-12B rank first and second overall; under math-adapted BPB they fall to fourth and seventh, while Qwen-3.5-35B-MoE and Qwen-3.5-9B rise from third and fifth to first and second. This is exactly the correction GSM8K accuracy already called for: both Gemma models are the GSM8K-weak family identified in Section 5.4, and math-domain adaptation demotes them specifically rather than reshuffling ranks arbitrarily.

Three general-text controls, and what they do and do not establish. The obvious alternative explanation is that any second corpus would have moved GSM8K. It would not: Table 9 shows Reddit at 0.691 and Hacker News at 0.791, both at or below the zero-shot baseline, against math’s 0.936. Two further checks address the remaining ways the result could be an artifact of the cohort. Math-adapted BPB is not simply a cleaner copy of the news-adapted ordering—the rank correlation between the two orderings is $\rho = 0.755$, well below the ~ 0.90 that would indicate a sharpened rather than reordered ranking, with individual models moving up to five ranks. And the improvement is not carried by one or two point movers: leave-one-out recomputation, dropping each of the 11 models in turn, stays in [0.915, 0.952], including when either Gemma model is dropped (tools/sanity_check_math_domain.py).

What the cohort size does not let us claim. None of these checks makes the difference statistically resolved, and we state that rather than let the leave-one-out result imply it. A paired bootstrap over the model cohort (20,000 resamples, seed and code in tools/analyze_alignment_matrix.py) puts the math-versus-news difference at $\Delta\rho = +0.209$ with a 95% interval of $[-0.096, +0.704]$, and the math-versus-zero-shot difference at +0.100 with $[+0.000, +0.393]$. Both intervals include or touch zero, because a rank correlation over 11 models cannot do better. The evidence here is the *pattern*—four adaptation corpora, one direction of effect, a reordering that demotes exactly the family

GSM8K already penalizes—and the pattern is what we ask the reader to weigh, not any single interval. Resolving it properly requires a larger cohort, which Section 6 names as the first thing we would add.

Scope. Two properties of the math corpus are worth stating precisely so the result is reproducible rather than oversold. First, the corpus is smaller than the other three domains (3,207 documents, 15.1 MB, versus 119,054 for news alone); the robustness checks above were run at this scale and hold at it, though they test sensitivity to which model is included, not to corpus size directly. Second, “domain-matched” here means topical match—arXiv research-paper prose—not GSM8K’s register of elementary word problems. That reflects a deliberate choice: the corpus was built to avoid answer-format leakage (Section 5.5’s contamination screen), not to mimic GSM8K’s surface form. Whether a closer-format corpus would push the effect further is the natural next experiment, not a gap in this one.

5.6 Selection Accuracy, with Uncertainty at the Model Level

Rank correlation over eleven models is a low-information summary, and pairwise selection accuracy—given two candidates, does lower BPB pick the better one?—uses the comparisons a practitioner actually makes. Each ranking induces 55 pairs. It is tempting to test those pairs as 55 independent trials; that is invalid, because they are 55 dependent comparisons among eleven models, and treating them as independent lets one model’s idiosyncrasy count as up to ten separate pieces of evidence. We therefore resample *models*: a cluster bootstrap over the eleven models, 20,000 draws, percentile intervals, the same unit of analysis as the bootstrap reported earlier.

The selection claim holds. Selection accuracy exceeds chance in 23 of the 24 (corpus, tier, benchmark) cells, the interval excluding 0.5 in every case but one. The strongest cells are news- and Hacker-News-adapted BPB against HellaSwag (0.964, CI [0.875, 1.000] and [0.840, 1.000]) and math-adapted BPB against GSM8K and MMLU-Pro (0.909, CI [0.755, 1.000] and [0.784, 1.000]). The single exception is math-corpus *zero-shot* BPB against HellaSwag (0.727, CI [0.478, 0.980]), the pairing with the least reason to work. This is the claim the leaderboard makes, and it survives testing at the unit the design randomises over.

The contrasts do not separate at this cohort size, and we do not claim they do. The difference in selection accuracy between math-adapted and news-adapted BPB is +0.127 on GSM8K (CI [−0.080, +0.367]), +0.127 on MMLU-Pro (CI [−0.060, +0.347]) and −0.127 on HellaSwag (CI [−0.362, +0.062]): the point estimates fall in the directions the matched-corpus hypothesis predicts, in opposite directions on the two benchmark families, and every interval contains zero. Adapting versus not adapting behaves the same way, +0.109 on GSM8K (CI [+0.000, +0.286]) and +0.164 on HellaSwag (CI [+0.000, +0.380]). Eleven models is not enough to separate two corpora on this statistic any more than on rank correlation, and switching statistics does not change that. What carries the steering claim is the pattern across *four* corpora reported in Section 5.5—three general-text corpora all at or below the do-nothing baseline on GSM8K, and only the topically matched one above it—not a significance test on any single pair of them.

Three robustness checks that need no additional compute. First, the steering hypothesis predicts a *joint* movement—toward the matched benchmark and away from the other—so the better-powered

statistic is a difference in differences: (GSM8K minus HellaSwag selection accuracy under math adaptation) minus the same quantity under a general-text corpus. It is +0.255 against news, +0.255 against Reddit and +0.218 against Hacker News, and +0.242 pooled.

The unit that statistic is resampled over decides what it means, and the obvious choice is wrong here. Resampling the eleven *models* gives strictly positive in 93.6% of 20,000 draws, exactly zero in 5.1% and negative in 1.3% (one-sided $p = 0.064$ counting every tie against the hypothesis, $p = 0.038$ under the mid- p convention, $p = 0.013$ counting none). But Section 5.1 argues that GSM8K rank movement is a *family* property—every Qwen model holds or improves, both Gemma models fall—and a bootstrap that treats family members as exchangeable draws contradicts the mechanism it is being used to support. Eleven models drawn from five publishers is an effective n well below eleven. We therefore report the family-clustered resampling as the primary reading:

resampling unit	clusters	positive	zero	negative	mid- p
model (naive)	11	93.6%	5.1%	1.3%	0.038
publisher $\times$ generation	6	85.7%	11.2%	3.1%	0.087
publisher family	5	80.9%	19.1%	0.0%	0.095

Two things follow, and they point in opposite directions; we state both rather than the convenient one. The *sign* is stable under the stricter clustering—not one of 20,000 publisher-level resamples is negative. The *magnitude* is less resolved than the model-level figure implied: under family clustering the effect no longer clears 0.05 on any tie convention ($p = 0.087$ and $p = 0.095$), where the model-level statistic did. These are not in tension, and neither rescues the other: a sign that never flips across 20,000 draws is evidence about direction only, and we do not describe the result as more robust overall on the strength of it. The mass at exactly zero is discreteness at five or six clusters, not degenerate draws.

A leave-one-family-out jackknife shows where the magnitude lives. Dropping the two Gemma models takes the pooled difference in differences from +0.242 to +0.056; dropping all six Qwen models takes it to +0.133, and dropping any of the three singleton families leaves it between +0.222 and +0.296. The steering effect is therefore concentrated in the family the paper already names as the one whose GSM8K rank falls under general-text adaptation, which is coherent with the mechanism but means the effect rests on two models.

That invites a fair charge of a double standard: Section 5.1 declines to report a gap-size correlation *because* one model’s deletion moves it, while this section reports a statistic that one family’s deletion moves by a comparable fraction. The asymmetry is deliberate and narrower than it looks. The correlation was a post-hoc addition with no pre-specified role, and a single coefficient is the only form it could take—so publishing it would have meant publishing a point estimate we could not defend. The difference in differences is the pre-specified primary statistic, and we report it with its full jackknife attached and make *no* significance claim for its magnitude. Neither is offered as a resolved number; the difference is that one is reported with its fragility in the same breath and the other could not have been. We report the direction as well supported and the magnitude as unresolved, and note that this is the same n problem as everywhere else in the study: the remedy is more *families*, not more models.

Second, corpus effects could in principle be confounded with optimisation randomness, since each cell is a single adaptation run. They are not, to the precision that matters here: re-running two Ministral-3-14B cells weeks later on different hardware reproduced zero-shot BPB exactly (0.7181

and 0.6718) and adapted BPB to within 0.0003 (0.6788 against a published 0.6791; 0.6221 exactly). Run-to-run variation is two orders of magnitude below the effects being compared.

Third, the benchmark rankings are treated above as ground truth, but they are themselves estimates. Recomputing selection accuracy while discarding model pairs whose benchmark gap is smaller than k combined standard errors—taken from the committed per-benchmark logs—leaves the result intact and slightly improves it: on GSM8K 0.909 at $k = 0$ becomes 0.933 at $k = 3$, and on HellaSwag 0.836 becomes 0.863. The errors the metric makes are concentrated in pairs the benchmark itself cannot separate, which is the failure mode one would want.

Against the obvious baseline: model size. A selection metric has to beat “pick the bigger model.” Over the full cohort the two tiers answer that differently, and the distinction matters. Parameter count orders 0.855 of pairs correctly on GSM8K and MMLU-Pro and 0.964 on HellaSwag. The free tier loses to it everywhere (0.836, 0.836, 0.800 for zero-shot news BPB). The paid tier beats it on both generative benchmarks (0.909 against 0.855) and loses on HellaSwag (0.836 against 0.964). So the honest headline is not that size wins, but that size is a strong and free baseline that only domain-matched adaptation improves on, and only on the benchmarks the adaptation corpus is matched to.

Even that flatters size for a structural reason: this cohort spans 0.5B to 35B, a $70\times$ range, so ordering by parameters is close to ordering by capability and the question is not one a practitioner needs help with.

The decision that is actually hard is between candidates of comparable size, and there the picture reverses. Restricted to pairs within $2\times$ of each other in parameters, size is barely better than a coin flip—0.545 on GSM8K and MMLU-Pro—while math-adapted BPB orders 0.818 of the same pairs correctly; within $1.5\times$, 0.714 against 0.857. HellaSwag runs the other way in every band (0.818 against 0.545 within $2\times$), which is the rotation of Section 5.5 appearing again: a metric steered toward mathematical prose orders reasoning better and commonsense worse. The same restriction prices the two tiers against each other, which the cost-fidelity frontier otherwise leaves qualitative. Within $2\times$, on GSM8K, picking the bigger model is right 0.545 of the time, zero-shot BPB 0.636, and math-adapted BPB 0.818: adaptation buys roughly 0.18 over doing nothing exactly where the choice is hard, against 0.07 over the whole cohort where size already answers the question. The value of the paid tier is concentrated in the close calls, which is also where a practitioner would be willing to pay for it.

Against the alternatives a practitioner actually weighs. Model size is the obvious baseline but not the only one. Table 10 sets out the options and what each one costs and tells you. The honest summary is that our method is not the best answer to every version of the question: when the deployment task is known and you can afford it, fine-tuning each candidate and measuring the task is strictly better evidence, and we say so. The protocol here is the right choice in the specific and common case where the deployment task is not yet a labelled benchmark, the candidates are close in size, and the corpus you care about is text you own.

These bands hold 7 to 20 pairs, so we report them as descriptive rather than tested; the claim they support is a scoping one. BPB earns its cost where size is uninformative, and the honest recommendation to a practitioner choosing between a 7B and a 9B model is different from the one we would give someone choosing between 0.5B and 35B, who does not need a metric at all.

Approach	Cost per candidate	What it does and does not tell you
Read published benchmark scores	free	Fast, and the right first look. Cannot separate a stronger model from a more specialised one, and cannot be audited: pre-training mixtures are undisclosed and the suites predate the models.
Run the suite yourself	hours of GPU	Removes cross-host and harness drift, which is worth up to 0.69 accuracy points here. Does not change <i>what</i> is measured, so it inherits every limitation above.
Arena / human preference	none, but indirect	Measures instructed chat behaviour. Base models are not what is rated, so it answers a different question than base-model selection.
Pick the larger model	free	As good as anything over a wide size range (0.855 here), and we recommend it there. Collapses to 0.545 among candidates within $2\times$, which is where the decision is actually hard.
Label a small in-house test set	human hours	The sharpest objection to this work, since Phase 1 already collects the data. Answers a task you must name in advance, which is the thing a new domain has not settled yet; and 500 labelled items give a binomial half-width near ± 4.4 points, wider than most gaps in this cohort. The same text yields $\sim 1.3M$ scored token predictions with no labelling at all. Where the task <i>is</i> nameable, this is the better instrument, and the automated form of it is our own next experiment.
Fine-tune and test on the task	highest	The strongest evidence available, and preferable whenever the task is known and labelled. Requires a labelled target task, which is exactly what a new domain usually lacks.
This protocol	minutes (free tier), ~ 1 GPU-h (adapted)	Uses text the evaluator owns and that post-dates every release, so it cannot have been prepared for. Reaches 0.818 among comparable-size candidates. Does not replace task evaluation, and cannot be claimed to be <i>more correct</i> than a benchmark where the two disagree.

Table 10: Base-model selection options. The protocol is proposed as a second, unpreparable axis for the case where the deployment task is not yet a labelled benchmark and the candidates are of comparable size—not as a replacement for task evaluation.

6 Reporting Standard and Limitations

The evidence supports a practical reporting contract and also fixes the boundary of the claim.

Version the full estimand. A leaderboard row is comparable only when corpus revision, split, context length, marker masking, risk screen, adaptation configuration, update budget, and BPB aggregation match. Any change creates a new benchmark ID. The public leaderboard validates these fields rather than accepting an isolated number.

Separate level, change, and uncertainty. Adapted BPB ranks endpoint fit; zero-shot→adapted reduction describes calibration. Task accuracies retain their own denominators and intervals. A large reduction, bold point estimate, or tiny numerical BPB gap should not be promoted into a stronger claim than the underlying comparison supports.

Scope of model selection. Adapted BPB measures base-model potential after the specified unsupervised domain intervention. Zero-shot BPB better describes immediate language-model fit, and a task benchmark remains the relevant criterion when the deployment task and elicitation protocol are known. The framework complements rather than replaces capability and safety evaluation.

Report both tiers, and choose between them on cost. Table 7 makes the practical recommendation two-tiered rather than singular. Zero-shot BPB requires one forward pass over held-out domain text and, in this cohort, already selects the generative-benchmark leader on three of four corpora. Adaptation costs a training run per candidate and does not dominate it: it buys alignment with whatever the adaptation corpus exercises, at the price of alignment elsewhere. A practitioner who will ship the base model as released should read the zero-shot column; one who will fine-tune on domain text should read the adapted column for *that* domain; and one who cares about a specific downstream capability should adapt on text matched to it, as Section 5.5 demonstrates for mathematics.

6.1 What a Fixed Update Budget Actually Measures

The fixed 250-step budget makes cells comparable, but Section 5.2 shows what is being compared, and it is narrower than “fit to the domain”. Across the two adapters studied there, roughly 83% of the gain falls on formatting, punctuation, and function words, with content words at 5.2% and 14.0%; the opening 32 positions of a block account for 23.0% of Gemma’s total gain. At 250 steps every model has consumed 8,000 blocks, which is 0.21 epochs of the news corpus. What that budget resolves best is therefore how quickly a model absorbs a corpus’s surface conventions—and the models that move most are exactly those that begin furthest from them (Gemma-4-12B 24.5% reduction, LFM2.5 36.4%, every Qwen between 3.6% and 5.4%).

This is a genuine limitation of the present estimand rather than a defect in the measurement, and it has a clean resolution we did not have the compute budget to run here: adapt every cell to its own convergence instead of to a shared step count. Concretely, replace “250 steps at one configuration” with “the minimum test BPB over a small *published* configuration grid, at the checkpoint that minimises validation BPB, with early stopping.” A fixed grid rather than an adaptive search matters for the same reason the fixed configuration replaced the sweep: an equal count of adaptive trials is not equal treatment, because the search path differs per model. Under that estimand the cheap surface-form gain is

exhausted for every candidate, unequal epoch counts across corpora become a property of the corpora rather than a confound, and steps-to-convergence becomes a second reportable axis—how expensive each base model is to bring to a domain—that the present protocol discards. We specify it here so the current leaderboard is read as what it is: fit after an equal and explicitly stated budget, not attainable fit.

6.2 Direct Measurement of Adapted Task Accuracy

Section 5.5 argues from rank movement that domain-matched adaptation trades general conditional-continuation calibration for task-relevant calibration. That is an inference about a mechanism, and the runs behind the published leaderboard did not retain adapters, so it had never been measured. We therefore repeated the news and mathematics adaptations retaining each adapter, merged them, and scored the adapted models on GSM8K and HellaSwag under the published task, few-shot, and extraction settings.

Two design points make the resulting deltas interpretable. First, the released weights were *re-scored on the same host and engine build* that produced the adapted numbers. This is not bookkeeping: the identical released weights scored on our earlier hardware and on this one differ by -0.60 to $+0.69$ accuracy points on GSM8K, which is larger than most of the effects being measured, while HellaSwag—a loglikelihood task—moves by at most 0.05 . Any adapted-versus-base delta that spans a hardware or engine change is uninterpretable at this scale, and cross-host comparison is common practice. Second, cells whose base score could not be re-measured on the same host are excluded from every aggregate rather than mixed in.

The ranking survives adaptation in the five models we could test. The claim in this subsection is scoped to that subset and should not be read as a property of the eleven-model leaderboard: the six excluded models are excluded by loader faults that correlate with architecture (a sparse MoE and multimodal-wrapped checkpoints), which is precisely the direction in which an adaptation-induced reordering would be most likely to appear. The unit here is a ranking, not a cell: four rankings of five models each—two corpora by two benchmarks—and *all four came back in exactly the base order*, Spearman $\rho = 1.00$ with not one inverted pair. A single such ranking arises by chance with probability $1/5! = 0.0083$, so one ranking already rejects at $p < 0.01$; the four are dependent, sharing both models and benchmarks, so we read them as one corroborated observation rather than four. It is a different measurement of the rank claim rather than an independent replication of it—the same cohort, the same adaptations—and it is evidence that these adaptation runs leave the ordering intact.

Absolute accuracy moves little, and by less than the cohort can localise. Mean changes are small and negative in all four cells, the largest being -2.82 points on GSM8K after mathematics adaptation. Two observations bound how much weight that can carry. The paired per-model contrast between the two corpora is -1.47 accuracy points with $t(4) = -1.63$, and only three of five models fall on the negative side, so the cohort cannot separate the corpora on accuracy the way it separates them on rank. And the natural mechanism—that degradation tracks how far a model moved in BPB—is carried entirely by one model: Pearson $r = -0.935$ over all cells and -0.036 once LFM2.5-1.2B is removed, that model reducing BPB by 36–42% where every other reduces it by 3.6–7.9%. A third check rules out the obvious artefact: answer-extraction drift is real, in that the lenient-minus-strict extraction gap widens under mathematics adaptation, but it cannot explain the change because the reported metric is already the lenient extractor.

Scope of this section. We pre-registered that no probe-versus-stress-test verdict would be issued until the full cohort was measured, and it was not: three Qwen3.5 and two Gemma-4 checkpoints are multimodal wrappers this inference stack cannot load, the latter because it refuses a configuration whose per-layer attention width genuinely varies (`head_dim = 256` against `global_head_dim = 512` over 40 sliding and 8 full attention layers), and Ministral-3-14B’s merged checkpoint requires a preprocessor configuration its publisher does not ship. All six exclusions were diagnosed before any excluded result existed, so their cause is independent of the outcomes, and each was refused a workaround that would have produced plausible numbers from a silently wrong model. We therefore report the accuracy comparison as directional and leave the mechanism question to the cohort extension of Section 6.1, which is the experiment that settles it: on the observed effect and spread—themselves imprecise at five models—separating the two corpora on accuracy would need a cohort on the order of fifteen, which is an indication of scale rather than a target. Section 5.5’s mechanism sentence stands as an inference, now explicitly labelled as one. The rank result above is unaffected—it is complete on the models measured.

The validation is circular, and here is how far that goes. The sharpest objection to this study is that it validates a metric said to be resistant to benchmark preparation by measuring its agreement with the benchmarks whose preparability motivates it. If those benchmarks are trustworthy enough to serve as ground truth, they are trustworthy enough to read directly; if they are not, agreement with them is agreement with noise. The objection is real and it bounds the claim rather than dissolving it. Our answer is that the two readings operate at different levels: preparation perturbs an individual model’s score on an individual suite, whereas what we validate against is a *cohort ordering* aggregated over eleven models and three benchmarks, which is a far more robust object than any single cell. Recovering that ordering from text no vendor could have seen establishes that the axis is real. What it cannot establish is that BPB is *more correct* than the benchmarks where the two disagree; on this evidence, disagreement is a flag to investigate, not a verdict in our favour. Settling that requires an outcome criterion independent of the static suites—in-domain cloze on held-out corpus splits is the design we have specified and not yet run—and until then the honest statement is that adapted BPB is an independent, unpreparable second axis that is as informative as size where the decision is hard, not a replacement for task evaluation.

Limitations. The protocol is designed to be honest about what it does not show, which is what makes the rest of it trustworthy.

Cohort size is the binding constraint. Eleven models is too few to resolve differences between rank correlations, as the bootstrap intervals in Section 5.5 show, and the family-level reading of the GSM8K residual rests on two Gemma models. Expanding the cohort is the single change that would most improve this work, and it is cheap, because the models that tighten the statistics are the small ones. The cohort is also English-centric and confounds architecture, data mixture, token budget, release generation, and parameter count.

The external yardstick is domain-general. We validate a domain-specific selection metric against MMLU-Pro, HellaSwag, and GSM8K, none of which is about news, Reddit, or Hacker News. When a domain-adapted ranking disagrees with a benchmark, this design cannot separate “the metric was wrong” from “the benchmark was the wrong yardstick for this domain”. Closing that gap requires an automatic, in-domain, non-likelihood task—masked-entity or numeric cloze built from the held-out split is the natural candidate, since it is uncontaminated by construction and scores exactly the content

positions Section 5.2 shows adaptation does *not* improve—and that experiment is not in this paper.

Measurement caveats. Fixed update and data budgets are not equal-FLOP budgets, and Section 6.1 states what the budget does and does not measure. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text (5.92–6.24 MB, a 5% spread) with block boundaries in different places; using the measured context sensitivity of Figure 4 to bound the resulting bias, it reaches at most $0.08\times$ the smallest adjacent rank gap and overturns no rank, but byte-aligned blocking would remove it. Adapters were not retained, so no adapted model is scored on a task benchmark anywhere in this work: the claim that domain-matched adaptation trades one kind of calibration for another is an inference from rank movement, not a direct measurement of accuracy before and after. Accuracy intervals cover test-set sampling only. The three diagnostic adapters and 64-window context study are explanatory probes, not cohort-wide replications. Reference-free contamination signals justify conservative removal but cannot verify proprietary training membership. Finally, the news text is third-party material: the release publishes its exact record-level fingerprint and verification tooling, but not the raw text without redistribution permission. These constraints limit causal and universal claims while leaving every reported comparison auditable.

7 Conclusion

The old choice between incomparable perplexities and narrow task accuracies is unnecessary. A recent raw-text snapshot, a byte denominator, conservative risk screening, and an explicitly stated adaptation budget turn predictive loss into an auditable base-model selection experiment.

What the experiment shows is not that adaptation makes the ranking better. Measured against external benchmarks, adaptation on general text makes it better on one axis and worse on the other: HellaSwag agreement rises from $\rho = 0.718$ to 0.982 on news while GSM8K falls from 0.836 to 0.727 and MMLU-Pro from 0.855 to 0.764, with the same pattern on Reddit and Hacker News. In selection terms, the free zero-shot ranking picks the GSM8K and MMLU-Pro leader outright and the adapted ranking gives up 2.3 and 4.2 accuracy points to pick the HellaSwag leader instead. Adaptation rotates the ranking toward whatever the adaptation corpus exercises.

That is a more useful result than a uniform improvement would have been, because it is controllable. Adapting on contamination-screened mathematical prose—not on GSM8K questions—puts GSM8K agreement at $\rho = 0.936$ with 50 of 55 model pairs ordered correctly, the best selection performance anywhere in this study, and restores the oracle pick; the same swap costs HellaSwag agreement, down to 0.818. Three general-text corpora, tried as controls, all leave GSM8K at or below the do-nothing baseline. The adaptation corpus is the instrument, and choosing it is the substantive decision the protocol asks of its user.

The recommendation is therefore a frontier rather than a number. Rank by zero-shot BPB on fresh target text when the model will be used as released—it is one forward pass and it is a strong general-purpose selector. Adapt and rank by adapted BPB when the model will be fine-tuned on domain text, reading the column for *that* domain. And when a specific downstream capability decides the choice, adapt on text matched to that capability and expect to trade breadth for it. Relative to train-before-test, the distinction is labels and estimand: that work harmonizes ranks after supervised task adaptation, whereas we compare off-the-shelf bases after unlabeled target-text preparation, across tokenizers.

We are explicit about what would change these conclusions, and about what turned out not to need changing. Eleven models cannot resolve a difference between two rank correlations, and our

bootstrap intervals say so rather than implying otherwise—but that is a limit of the statistic, not of the evidence. Section 5.6 reports the more informative statistic—selection accuracy over model pairs—with uncertainty still taken at the model level, and the selection claim holds in 23 of 24 cells while the between-corpus contrast remains unresolved. Testing those pairs as independent trials would make the contrast significant; it would also be wrong, and we say so rather than banking it. Three limits remain. The external yardsticks are domain-general while the metric is domain-specific, and a fixed step budget measures how fast a model absorbs a corpus’s surface form rather than how well it can ultimately model the domain. Each has a concrete remedy—a wider cohort, an in-domain generative task, adaptation run to convergence—and the cohort is the one that would settle the between-corpus contrast. The released code, manifests, and per-cell artifacts are arranged so that anyone, including us, can run all of them against exactly these numbers.

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